MA3521 — Separation of Variables

Name:

Problem 1 $y' = e^{t+y^2}/y, \qquad y(1) = 2$

1a. Identify the factors in $y' = g(t)h(y)$.

$g(t) =$ $\qquad$ $h(y) =$

1b. Write the integrals. Do not evaluate.

$\int$ $dy = \int$ $dt + C$

1c. Evaluate the integrals.

$=$ $+ C$

1d. Evaluate the constant.

$C =$

Problem 2 $y' = e^{t+y}, \qquad y(0) = 0$

2a. Identify the factors in $y' = g(t)h(y)$.

$g(t) =$ $\boxed{}$ $\qquad h(y) =$ $\boxed{}$

2b. Write the integrals. Do not evaluate.

$\int \boxed{}\, dy = \int \boxed{}\, dt + C$

2c. Evaluate the integrals.

$\boxed{} = \boxed{} + C$

2d. Evaluate the constant.

$C =$ $\boxed{}$